WORKED SOLUTION

STEP III 2005, Question 13

Probability and Statistics • Geometric Distribution

STEP III 2005, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q13

The key idea

The cards from $w+1$ to n do nothing at all: they pay nothing and they don't end the game. So in part (i) we can simply ignore them and watch only the draws that show $0, 1, \ldots, w$; each of those is a fair pick from $w+1$ equally likely numbers, and the total win is then a geometric random variable. In part (ii), where the draws are no longer independent, we split the total win into a sum of indicators, one for each paying card, and use linearity of expectation.

Solution

Call a draw *relevant* if it shows one of $0, 1, \ldots, w$, and *irrelevant* if it shows one of $w+1, \ldots, n$.

Part (i)

Why the two events have the same probability. An irrelevant draw changes nothing: the player receives nothing and the game carries on. So the player's total depends only on the sequence of relevant draws: each non-zero relevant draw earns £1, and the first relevant draw showing 0 ends the game. Hence the player wins exactly £3 precisely when, among the relevant draws, the first three show non-zero numbers and the fourth shows 0. These are the same event, so they have the same probability.

(Each draw is relevant with probability $\frac{w+1}{n+1} > 0$, independently, so with probability 1 there are infinitely many relevant draws and in particular a first, second, third and fourth one; the game ends with probability 1.)

Ask yourself Why is each relevant draw equally likely to show any of $0, 1, \ldots, w$, regardless of what came before?

Because the card is replaced each time, the draws are independent, each uniform on $\{0, 1, \ldots, n\}$. Given that a particular draw is relevant, it is uniform on $\{0, 1, \ldots, w\}$, since

those $w+1$ outcomes were equally likely to begin with. Independence of the draws means the values of successive relevant draws are also independent. So each relevant draw is 0 with probability $\frac{1}{w+1}$ and non-zero with probability $\frac{w}{w+1}$, independently of the others.

Therefore

$$P(\text{win exactly £3}) = \left(\frac{w}{w+1}\right)^3 \cdot \frac{1}{w+1} = \frac{w^3}{(w+1)^4},$$

as required.

General r. Exactly the same argument with r in place of 3 (the first r relevant draws non-zero, the $(r+1)$th zero) gives

$$P(\text{win exactly £}r) = \frac{w^r}{(w+1)^{r+1}}, \qquad r = 0, 1, 2, \ldots$$

Writing $q = \frac{w}{w+1}$, this is $(1-q)q^r$, a geometric distribution; these probabilities sum to 1 as they should.

Expected total win. Let X be the total win in pounds. Differentiating $\sum_{r\geq 0} q^r = \frac{1}{1-q}$ for $|q| < 1$ and multiplying by q gives $\sum_{r\geq 0} rq^r = \frac{q}{(1-q)^2}$. So

$$E(X) = \sum_{r=0}^{\infty} r(1-q)q^r = (1-q)\cdot\frac{q}{(1-q)^2} = \frac{q}{1-q} = \frac{w/(w+1)}{1/(w+1)} = w.$$

The expected total win is £w.

Part (ii)

Now the cards are not replaced, so the game is the same as shuffling the pack into a uniformly random order and turning cards over until 0 appears. The player receives £1 for each of the cards $1, \ldots, w$ that comes before card 0 in this order (every card before 0 is drawn, and none after it).

For $k = 1, \ldots, w$ let $I_k = 1$ if card k comes before card 0, and $I_k = 0$ otherwise. The total win is $Y = I_1 + I_2 + \cdots + I_w$.

> **Ask yourself** The I_k are not independent. Does that matter?

No: the expectation of a sum is the sum of the expectations whatever the dependence. So we need only $P(I_k = 1)$. In a uniformly random order, swapping the positions of cards k and 0 is a bijection between orders with k before 0 and orders with 0 before k, so the two are equally likely and $P(I_k = 1) = \frac{1}{2}$. Hence

$$E(Y) = \sum_{k=1}^{w} E(I_k) = \sum_{k=1}^{w} \tfrac{1}{2} = \frac{w}{2},$$

which is half of the expected win w in part (i), as required.

Checks

- With $w = n$ (every non-zero card pays) and no replacement, the position of card 0 is uniform on $1, \ldots, n+1$, so the mean number of cards before it is $\frac{n}{2}$, matching $\frac{w}{2}$.
- With replacement and $w = 1$, each relevant draw is a fair coin between 0 and 1, so $P(X = r) = 2^{-(r+1)}$ and $E(X) = 1$, matching $w^r/(w+1)^{r+1}$ and $E(X) = w$.
- A simulation of 2×10^5 games with $n = 5$, $w = 2$ gave $P(X = 3) \approx 0.0995$ against $\frac{8}{81} \approx 0.0988$, and a mean win of 1.995 against 2. Exact enumeration of all orders for small packs (for example $n = 5$, $w = 3$) gives mean $\frac{3}{2} = \frac{w}{2}$ without replacement.

Common slips

- Using $\frac{1}{n+1}$ for the chance that a draw is 0. That is the chance per draw, but irrelevant draws can occur any number of times in between; conditioning on relevant draws is what gives the clean $\frac{1}{w+1}$.
- Forgetting $r = 0$: the first relevant draw may be 0, giving probability $\frac{1}{w+1}$ of winning nothing.
- Trying to find the full distribution of the win in part (ii). It isn't needed; indicators and linearity do the job in two lines.

Beyond STEP

Ignoring the irrelevant draws is an example of *thinning*, or looking at a process only at the times something interesting happens; it reappears with Markov chains (the "jump chain") and Poisson processes. Writing a count as a sum of indicators and using linearity of expectation, even when the indicators are dependent, is one of the most useful tricks in first-year probability and combinatorics. Both ideas are developed in Blitzstein and Hwang, *Introduction to Probability*, and in Grimmett and Welsh, *Probability: An Introduction*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.